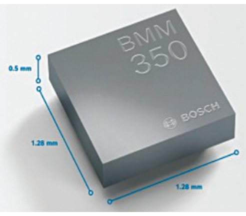
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Magnetometer with field shock recovery

The 16-bit, 3-axis magnetometer BMM350 launched by BOSCH utilises innovative tunnel magnetoresistance (TMR) technology and features a distinctive field shock recovery capability.

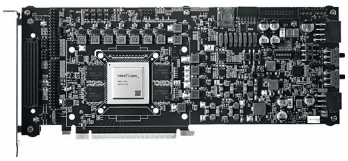


By incorporating this feature, the device exhibits exceptional resilience against external magnetic fields, guaranteeing consistent high accuracy in all circumstances. With compact dimensions of 1.28mm x 1.28mm x 0.5mm, this device is nearly invisible. It is best suited for wearables, hearables, smartphones, tablets, augmented reality (AR) and virtual reality (VR) devices, and various types of vehicles. The magnetometer offers a remarkable 20-fold reduction in current consumption compared to its previous generation. The built-in MEMS sensors allow users to operate smart devices through gestures or voice input. The sensor has high sensitivity and resolution, allowing it to detect even subtle changes in the magnetic field.

BOSCH
https://www.bosch.com

Versatile, energy-efficient AI SoC

ATOM is an artificial intelligence (AI) enabled system on chip (SoC) launched by Korean AI startup Rebellions that bridges latency-critical machine learning (ML) tasks and hyper-scale ML services. According to the company, the SoC is a high-speed inference machine ideal for training models used in generative AI applications such as ChatGPT. The chip supports cutting-edge AI networks, including state-of-the-art GPT language models, with top performance and energy efficiency. ATOM allows simultaneous support for up to 16 isolated jobs at different hardware and

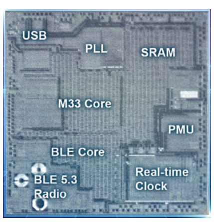


software levels through efficient partitioning. Maximised energy efficiency is achieved by implementing a low-power design methodology and utilising tidy core structures. The AI core can accommodate networks with different depths and complexities, ranging from small-scale applications to high-level data centre services.

Rebellions Inc.
https://rebellions.ai

22nm wireless microcontroller

Renesas has announced its first wireless microcontroller (MCU) utilising advanced 22nm process technology. The MCU provides an innovative solution to customers

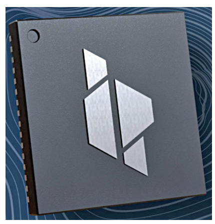


by combining Bluetooth 5.3 Low Energy (LE) with a built-in software-defined radio (SDR). Whether in the development phase or after deployment, these devices can be easily upgraded with new application software or enhanced Bluetooth features, ensuring adherence to the latest specification versions and offering a future-proof solution. End product manufacturers can use the wireless MCU to harness the features from previous Bluetooth LE specification releases. The processor uses a smaller die area for the same functionality, resulting in smaller chips with higher integration of peripherals and memory. It paves the path for future-generation devices that assist customers in future-proofing their designs and guaranteeing long-term availability.

Renesas Electronics Corporation
https://www.renesas.com/in/en

Digital-to-RF chip for 4G/5G O-RAN

Innophase has launched Hermes TWO (H2), a digital to radio frequency (RF) radio for 4G/5G open radio access network (O-RAN) radio units. The product



distinguishes itself from similar products in the market by incorporating a patented digital-to-RF radio architecture and digital processing elements into a single-chip design, significantly reducing power consumption. The device utilises advanced CMOS technology, providing standardisation and enhancing system integration and efficiency. It offers a simplified, low-power system architecture suitable for various cellular network radios. H2 introduces a revolutionary architecture that directly converts digital signals to an impressive 27dBm RF power output to achieve an extended wireless range. The device finds applications in various areas, including in-building wireless applications or private networks, network densification, and high-efficiency cells.

Innophase
https://innophaseinc.com